R Programming Practical Examination

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Part A: basic Operations

A1 Create v as the sequence 5, 10, 15, . . . , 50 using seq(). Print its length, sume and mean.

```
v <- seq(5, 50, by=5)
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2 Create w that repeats the vector c(1, 2, 3) four times, and z that repeats each element of c (1,2, 3) four times. Print both.

```
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3 Compute and print: the remainder of 17 divided by 5, the integer division of 17 by 5, 2 to the power of 10, the square root of 144, and pi rounded to 3 decimals.

```
17 %% 5
```

```
## [1] 2
```

```
17 %/% 5
```

```
## [1] 3
```

```
2^10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```
round(pi, 3)
```

```
## [1] 3.142
```

A4 Given a <- c(2, 4, 6, 8) and b<- c(1,3,5,7), print a + b, a * b the logical comparison a > b, and the dot product sum (a * b) .

```
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

a + b
```

```
## [1]  3  7 11 15
```

```
a * b
```

```
## [1]  2 12 30 56
```

```
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a * b)
```

```
## [1] 100
```

A5 Use paste() to produce “Student-1” “Student-2” “Student-3”. Then identifying the datatype of 42, 42L, “42”, 4li, and TRUE. What are its output?

```
paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```
class(42)
```

```
## [1] "numeric"
```

```
class(42L)
```

```
## [1] "integer"
```

```
class("42")
```

```
## [1] "character"
```

```
class(411)
```

```
## [1] "numeric"
```

```
class(TRUE)
```

```
## [1] "logical"
```

```
The output for the paste("Student", 1:3, sep = "-") is "Student-1" "Student-2" "Student-3", for the cla
```

Part B: Data Frames and Conditional Statements

B1 Create a data frame students with 8 rows and these columns (use stringsAsFactors = FALSE; sex is “M” or “F”):

```
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
```

```
)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2 Display the structure, dimensions, colums names, and summary of students.

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00  
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25  
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50  
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75  
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00  
##  Max.   :8.00                                            Max.   :95.00  
##      final         attendance   
##  Min.   :65.00   Min.   : 60.00  
##  1st Qu.:77.00   1st Qu.: 81.75  
##  Median :81.00   Median : 89.00  
##  Mean   :82.12   Mean   : 86.00  
##  3rd Qu.:91.50   3rd Qu.: 95.75  
##  Max.   :96.00   Max.   :100.00
```

B3 Add a column average = mean of midterm and final for each student, rounded to 1 decimal.

```
students$average <- round(rowMeans(students[, c("midterm", "final")]), 1)
students
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
```

```
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

B4 Add a logical column passed that is TRUE when average >= 70 and attendance >= 75.

```
students$passed <- students$average >= 72 & students$attendance >= 75
students
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

B5 Using ifelse(), add a column standing with these values based on average: * “Honors” if average >= 90 * “Good” if average >= 75 (and below) * “Needs Support” otherwise

```
students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs Support"))
students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

B6 Write a function classifying(score) that uses an if/ elde if/ else/ chain and returns “Distinction” for 85 or above, “Merit” for 70 to 84. “Pass” for 50 to 69, and “Fail” below 50. Test it with sapply (c(90, 73, 55, 40), classify) and print the result.

```
classify <- function(score) {
  if (score >= 85) {
    return("Distinction")
  } else if (score >= 70) {
    return("Merit")
  } else if (score >= 50) {
    return("Pass")
  } else {
    return("Fail")
  }
}

sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1 Create a data frame new_students with two students. Its columns must match the original, including average, passed and standing, computed the same way:

- id 9, Ines, F, section C, midterm 81, final 84, attendance 92
- id 10, Jamal, M, section A, midterm 70, final 68, attendance 80

Use rbind() to append them to students, saving the result as students_all.

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- round(rowMeans(new_students[, c("midterm", "final")]), 1)
new_students$passed <- new_students$average >= 72 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                         ifelse(new_students$average >= 75, "Good", "Needs Support"))

students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

C2 Create the numeric vector quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6). Use cbind() to attach it to students_all as a column named quiz. Confirm the result is still a data frame with class().

```
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz)
class(students_all)
```

```
## [1] "data.frame"
```

```
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
```

```
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
##    quiz
## 1     8
## 2     7
## 3    10
## 4     6
## 5     7
## 6     9
## 7    10
## 8     5
## 9     8
## 10    6
```

C3 Create the matrix m <- cbind(midterm = students_all$midterm, final = students_all$final) and use rbind() to append a final row containing the column means, labelled "mean".

```
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m_with_mean <- rbind(m, mean = colMeans(m))
m_with_mean
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

C4 In a comment (2-3 sentences), explain what error rbind() produces if the data frames have different column names, and show the output if you safely catch it with tryCatch() tryCatch(rbind(students, bad), error = function(e) { message("rbind failed:", conditionMessage(e));

```
bad <- data.frame(id = 11, student_name = "Katy") # Incorrect column name

tryCatch(
  rbind(students, bad),
  error = function(e) {
    message("rbind failed: ", conditionMessage(e))
    NULL
  }
)
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```

```
The rbind() requires all data frames that is being combined to have matching column names and structures
```

Part D: Subsetting

Use students_all (with quiz) for all questions. Print every result. D1 Extract the name column in three ways: $, [[“name”]], and [, “name”].

```
students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2 Extract rows 2 to 5 and only the columns name, midterm, final.

```
students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3 Using logical subsetting, select students withattendance’ above 90.

```
students_all[students_all$attendance > 90, ]
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

D4 Using subset(), select students of sex “F” in section “A” or “C” whose final is at least 75; return only name, section and final.

```
subset(
  students_all,
  sex == "F" & (section == "A" | section == "C") & final >= 75,
  select = c("name", "section", "final")
)
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

D5 Use which.max() and which.min() to report the name of the student with the highest and lowest average.

```
# For the student with the highest average
students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```
# For the student with the lowest average
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

D6 Sort students_all by average in descending order, then print the top 3 rows.

```
students_sorted <- students_all[order(-students_all$average), ]
head(students_sorted, 3)
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic Factors

E1 Convert section and sex in students_all to factors with factor(). Print levels() of each.

```
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)

levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

E2 Use table() to count students per sex and per section. *Try the two-way table of sex by section. table(students_all$sex, $students_all$section). What is the output?

```
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

```
The outputs are the 3 colums A, B and C and the 2 rows F and M. In the first row and first column the ou
```

E3 Print as. integer() of the sex factor. In a comment, explain what the integers represent.

```
as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```
# Underneath factor labels, R stores levels as integer values corresponding to alphabetical order of th
```

E4 Create a new factor sex_label from sex using factor() with levels = c("F", "M") and labels = c("Female", "Male"). Add it to students_all and print its levels().

```
sex_label <- factor(students_all$sex, levels = c("F", "M"), labels = c("Female", "Male"))
students_all$sex_label <- sex_label
levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

E5 Print summary() of sex_label. Then, using logical subsetting and mean(), print the mean average for each level of sex_label.

```r
summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```r
# For the mean average for Females
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```r
# For the mean average for Males
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 79.3
```